Interestingly, binary search is a realization of making the information-theoretically optimal decision in this game of searching for the location of a number within a sorted list, if we were to tackle the problem from a probabilistic angle.

First, let's use the random variable L to denote the location of our target z within the sorted list. Here, we would like to use a distribution to describe our belief about the variable. Since from our perspective, any location within the list is equally likely to contain the value of z, we will use a uniform distribution for modeling.

Figure 6.6 visualizes this distribution, which again represents our belief about the location of z. Since each location is equally likely as any other, the probability that a given location contains z is uniformly one divided by one hundred, or one percent.

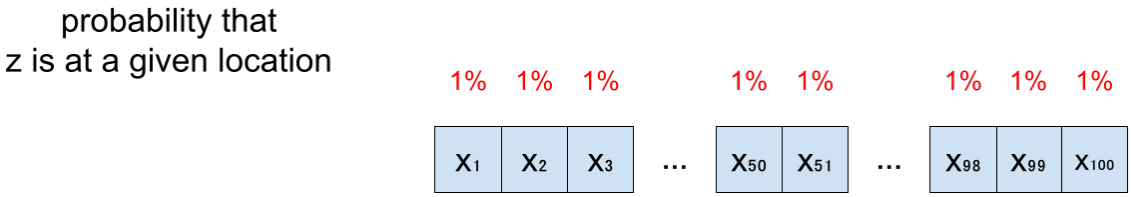


Figure 6.6 Prior distribution of the location of target z within the 100-element list. Since each location is equally likely as any other, the probability that a given location contains z is one percent.

NOTE **Entropy of a uniform distribution**

Let's try computing the entropy of this uniform distribution. Remember the formula for the entropy is p log p, where we sum over the different possible events indexed by i. This is equal to:

$$-\sum_{i=1}^{100} 0.01 \log 0.01 \approx 6.64$$

So, the amount of uncertainty we have in our prior distribution of L is roughly 6.64.

We tackle the same question: how should we search over this 100-element list in order to locate z as quickly as possible? We do this by following the entropy search procedure described in

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